

Retail Sales & Returns Analytics using SQL Server

An End-to-End SQL Analytics Project for Business Intelligence and
Decision Support

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1. Project Overview

This project was developed to simulate a real-world retail analytics scenario using Microsoft SQL Server. The primary objective was to transform raw transactional sales and return data into a structured analytical reporting layer capable of supporting business intelligence and decision-making. Instead of focusing only on SQL syntax, the project emphasized solving real business problems by designing reusable SQL views, defining meaningful KPIs, and answering critical business questions across sales performance and return analysis.

During the project, reusable analytical views were created to evaluate product performance, category performance, customer performance, product returns, category returns, and return reasons. Business KPIs such as revenue, profit, quantity sold, return rate, and ranking metrics were developed to provide meaningful insights into overall business performance. The final reporting layer was specifically designed to serve

as a backend data source for future Power BI dashboards, creating an end-to-end business intelligence solution.

2. Business Problem

The retail company aims to better understand its sales performance and product return behavior to improve profitability, identify high-performing products, categories, and customers, and reduce product returns through data-driven decision-making.

3. Project Objectives

The primary objectives of this project were to:

- Analyze retail sales performance across products, categories, and customers.
- Evaluate customer purchasing behavior through transactional sales analysis.
- Measure product and category revenue, profitability, and sales performance using business KPIs.
- Analyze product return behavior and identify the primary reasons contributing to returns.
- Design reusable SQL views to support scalable reporting and business analysis.
- Develop a structured reporting layer for seamless integration with Power BI dashboards.

5. Technology Stack

Category	Technology
Database	Microsoft SQL Server
Language	T-SQL
IDE	SQL Server Management Studio
Visualization	Power BI (Planned)

6. Dataset Overview

The project utilizes a retail transactional database consisting of six interconnected tables that capture customer information, product catalog details, sales transactions, and product return records. Together, these tables provide the foundation for analyzing sales performance, customer purchasing behavior, profitability, and return trends.

Table	Description	Primary Key
Customers	Stores customer information such as customer ID, name, contact details, and location.	customer_id
orders	Contains order-level information including customer, order date, and order status.	order_id
order_items	Stores individual products purchased within each order, including quantity and transaction price.	order_item_id

Products	Product master table containing selling price, cost price, and category information.	product_id
Categories	Defines product categories used for sales and return analysis.	category_id
returns	Records returned products along with return dates and return reasons.	return_id

Dataset Relationships

The dataset follows a relational database structure where customers place orders, each order consists of one or more order items, order items reference products, products belong to categories, and returned products are recorded in the returns table. These relationships enable comprehensive sales and return analysis across multiple business dimensions.

7. Data Model

The retail database follows a relational data model where customer transactions are recorded across multiple interconnected tables. These relationships enable the analysis of sales performance, customer purchasing behavior, product profitability, and product returns. The data model serves as the foundation for building reusable analytical SQL views and business KPIs.

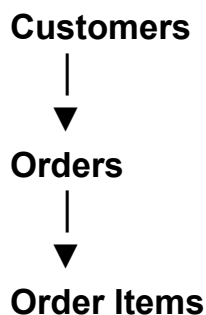




Figure 1. Retail Data Model

The data flows from customers placing orders to individual products being purchased. Products are grouped into categories, while returned items are tracked separately through the returns table. This relational structure forms the foundation for all sales, customer, category, and return analyses performed in this project.

8. Business Questions

The project was designed to answer key business questions related to sales performance, customer behavior, profitability, and product returns. These questions guided the development of the SQL views and business KPIs throughout the project.

Sales Performance Analysis

Product Performance

- Which products generate the highest revenue?
- Which products generate the highest profit?
- Which products have the highest sales volume?

Category Performance

- Which product categories generate the highest revenue?
 - Which product categories generate the highest profit?
 - Which categories sell the highest quantity of products?
-

Customer Performance

- Which customers generate the highest revenue?
 - Which customers generate the highest profit?
 - Which customers purchase the highest quantity of products?
-

Return Analysis

Product Returns

- Which products are returned most frequently?
 - Which products have the highest return rate?
-

Category Returns

- Which categories experience the highest number of returns?
 - Which categories have the highest return rate?
-

Return Reasons

- What are the most common reasons for product returns?
- What percentage of total returns does each return reason contribute?

9. SQL Views & KPIs

Introduction

To support business reporting and simplify future dashboard development, a series of reusable SQL views were created. Each view focuses on a specific business area and provides pre-calculated KPIs that can be directly consumed by reporting tools such as Power BI.

9.1 vw_product_performance

Purpose

Evaluates the sales performance of individual products by calculating key business metrics and ranking products based on revenue and profitability.

KPIs

- Total Quantity Sold
- Total Revenue
- Total Profit
- Revenue Rank
- Profit Rank

Results Messages								
	product_id	product_name	category_name	total_quantity_sold	total_Revenue	total_Profit	revenue_rank	profit_rank
1	1	iPhone 14	Electronics	201	15075000.00	3015000.00	1	1
2	3	Dell Laptop	Electronics	150	9000000.00	2250000.00	3	2
3	2	Samsung Galaxy S23	Electronics	147	9555000.00	2205000.00	2	3
4	8	Sofa Set	Home	212	5300000.00	2120000.00	4	4
5	9	Dining Table	Home	175	3500000.00	1400000.00	5	5
6	14	Cricket Bat	Sports	212	848000.00	530000.00	6	6
7	10	Mixer Grinder	Home	187	841500.00	467500.00	7	7
8	12	Perfume	Beauty	224	560000.00	380800.00	10	8
9	7	Running Shoes	Fashion	196	588000.00	352800.00	9	9
10	4	Bluetooth Headphones	Electronics	188	658000.00	282000.00	8	10
11	6	Women Dress	Fashion	209	376200.00	229900.00	11	11
12	15	Football	Sports	197	295500.00	197000.00	12	12
13	13	Yoga Mat	Sports	225	270000.00	180000.00	13	13
14	19	Toy Car	Toys	183	164700.00	109800.00	15	14
15	18	Puzzle Game	Toys	182	145600.00	109200.00	16	15

9.2 vw_category_performance

Purpose

Aggregates product-level data to evaluate overall category performance and compare categories using business KPIs.

KPIs

- Total Quantity Sold
- Total Revenue
- Total Profit
- Revenue Rank
- Profit Rank

Results		Messages					
	category_name	total_quantity_sold	total_revenue	total_profit	revenue_rank	profit_rank	
1	Electronics	686	34288000.00	7752000.00	1	1	
2	Home	574	9641500.00	3987500.00	2	2	
3	Sports	634	1413500.00	907000.00	3	3	
4	Fashion	578	1102600.00	669200.00	4	4	
5	Beauty	395	679700.00	466300.00	5	5	
6	Toys	365	310300.00	219000.00	7	6	
7	Grocery	395	361800.00	160100.00	6	7	
8	Books	398	165700.00	117100.00	8	8	

9.3 vw_customer_performance

Purpose

Analyzes customer purchasing behavior by measuring order activity, purchasing volume, revenue contribution, and profitability.

KPIs

- Total Orders
- Total Quantity Purchased
- Total Revenue
- Total Profit
- Revenue Rank
- Profit Rank11

Results		Messages						
	customer_id	customer_full_name	customer_total_orders	customer_total_quantity_purchased	customer_total_revenue	customer_total_profit	customer_revenue_rank	customer_profit_rank
1	25	Customer25 Last25	5	43	987450.00	248950.00	1	1
2	45	Customer45 Last45	8	47	685300.00	213800.00	6	2
3	125	Customer125 Last125	9	49	781100.00	207500.00	2	3
4	22	Customer22 Last22	5	31	729000.00	198200.00	3	4
5	116	Customer116 Last116	8	56	638500.00	198200.00	12	4
6	6	Customer6 Last6	5	43	662350.00	188750.00	9	5
7	91	Customer91 Last91	7	50	537600.00	183650.00	22	6
8	107	Customer107 Last107	5	34	667600.00	180650.00	8	7
9	199	Customer199 Last199	7	43	553050.00	176750.00	20	8
10	108	Customer108 Last108	5	35	650000.00	175800.00	10	9
11	104	Customer104 Last104	4	22	722000.00	173400.00	4	10
12	144	Customer144 Last144	6	46	646800.00	173050.00	11	11
13	47	Customer47 Last47	6	23	694200.00	169400.00	5	12
14	48	Customer48 Last48	5	31	674200.00	168650.00	7	13
15	161	Customer161 Last161	6	35	601300.00	168500.00	13	14

9.4 vw_return_details

Purpose

Serves as the base analytical view for all return-related analysis by combining customer, order, product, category, and return information into a single reporting dataset.

Contains

- Customer Information
- Product Information
- Category Information
- Return Date
- Return Reason

Results		Messages							
	return_id	customer_id	customer_name	product_id	product_name	category_id	category_name	return_date	return_reason
1	67	169	Customer169 Last169	13	Yoga Mat	5	Sports	2024-04-23	Wrong Product
2	14	167	Customer167 Last167	5	Men T-Shirt	2	Fashion	2025-04-11	Size Issue
3	21	22	Customer22 Last22	19	Toy Car	7	Toys	2024-05-17	Damaged Item
4	20	113	Customer113 Last113	1	iPhone 14	1	Electronics	2025-06-17	Size Issue
5	17	184	Customer184 Last184	11	Face Cream	4	Beauty	2025-01-10	Size Issue
6	31	46	Customer46 Last46	3	Dell Laptop	1	Electronics	2024-06-26	Wrong Product
7	10	117	Customer117 Last117	18	Puzzle Game	7	Toys	2025-12-02	Wrong Product
8	63	179	Customer179 Last179	14	Cricket Bat	5	Sports	2025-02-07	Damaged Item
9	74	23	Customer23 Last23	18	Puzzle Game	7	Toys	2024-12-01	Size Issue
10	51	101	Customer101 Last101	20	Rice 5kg	8	Grocery	2025-05-02	Damaged Item

9.5 vw_product_return_analysis

Purpose

Measures product return performance by calculating the number of returns, return rate, and ranking products based on return frequency.

KPIs

- Total Returns
- Return Rate
- Return Rank

Results		Messages					
	product_id	product_name	category_name	total_quantity_sold	total_returns	return_rate	return_rank
1	18	Puzzle Game	Toys	182	2	1.10	1
2	19	Toy Car	Toys	183	1	0.55	2
3	20	Rice 5kg	Grocery	187	1	0.53	2
4	1	iPhone 14	Electronics	201	1	0.50	2
5	3	Dell Laptop	Electronics	150	1	0.67	2
6	5	Men T-Shirt	Fashion	173	1	0.58	2
7	11	Face Cream	Beauty	171	1	0.58	2
8	13	Yoga Mat	Sports	225	1	0.44	2
9	14	Cricket Bat	Sports	212	1	0.47	2

9.6 vw_category_return_analysis

Purpose

Summarizes return performance at the category level to identify categories with the highest return volume and return rate.

KPIs

- Total Returns
- Return Rate
- Return Rank

Results		Messages			
	category_name	total_quantity_sold_per_category	total_returns_per_category	return_rate_per_category	return_rank
1	Toys	365	3	0.82	1
2	Electronics	351	2	0.57	2
3	Sports	437	2	0.46	2
4	Beauty	171	1	0.58	3
5	Fashion	173	1	0.58	3
6	Grocery	187	1	0.53	3

9.7 vw_return_reason_analysis

Purpose

Analyzes the distribution of return reasons to understand the primary factors contributing to product returns.

KPIs

- Total Returns
- Return Percentage
- Return Rank

Results		Messages		
	return_reason	total_returns	return_percentage_of_total	return_rank
1	Wrong Product	34	34.00	1
2	Damaged Item	33	33.00	2
3	Size Issue	33	33.00	2

10. Business Insights

The analytical views developed throughout this project were used to derive meaningful business insights related to sales performance, customer behavior, profitability, and product returns. These findings can support strategic decision-making aimed at improving revenue, profitability, and customer satisfaction.

Sales Performance Insights

Product Performance

- Identified the highest-performing products based on **total revenue** and **total profit**.
- Ranked products using revenue and profit KPIs to support product-level performance evaluation.
- Measured product sales volume using total quantity sold.

Category Performance

- Evaluated category performance using revenue, profit, and quantity sold.
- Identified top-performing categories based on overall business contribution.
- Enabled category-level comparisons through reusable ranking metrics.

Customer Performance

- Identified the highest-value customers based on revenue generation and profitability.
 - Measured customer purchasing behavior through total orders and total quantity purchased.
 - Ranked customers to support customer segmentation and future loyalty initiatives.
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Return Analysis Insights

Product Returns

- Calculated return rates using **returned quantity relative to total quantity sold**, providing a more meaningful KPI than raw return counts.
 - Identified products with the highest number of returns.
 - Ranked products based on return frequency to highlight potential quality or fulfillment issues.
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Category Returns

- Aggregated product return data to evaluate return performance across categories.
 - Calculated category-level return rates to normalize comparisons across different sales volumes.
 - Identified categories with relatively higher return behavior.
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Return Reasons

- Analyzed the distribution of return reasons across all returned products.
- Found that return reasons were nearly evenly distributed:
 - **Wrong Product – 34%**
 - **Damaged Item – 33%**
 - **Size Issue – 33%**

- The balanced distribution suggests that product returns are influenced by multiple operational factors rather than a single dominant issue.
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Business Impact

This is the section that most portfolio projects **don't have**, but I think yours should.

The analytical reporting layer developed in this project enables business users to monitor sales performance, customer value, profitability, and return behavior through reusable SQL views. These insights can support better inventory planning, product quality monitoring, customer relationship management, and operational improvements aimed at reducing return rates and improving overall business performance.

12. Skills Demonstrated

The project demonstrates the application of technical SQL skills, business analysis techniques, and analytical thinking to solve real-world retail business problems.

Category	Skills Demonstrated
SQL	SELECT, INNER JOIN, GROUP BY, Aggregate Functions, Window Functions (DENSE_RANK), Views
Database	Relational Database Design, Data Modeling, Reporting Layer Development
Business Analysis	KPI Design, Sales Analysis, Customer Analysis, Return Analysis, Performance Ranking
Data Analytics	Revenue Analysis, Profit Analysis, Return Rate Calculation, Trend Identification, Business Insights

Problem Solving

Analytical Thinking, KPI Development, Query Optimization, Reusable SQL View Design

13. Key Learnings

This project provided valuable experience in designing analytical SQL solutions and reinforced the importance of thinking beyond query writing. Some of the key learnings include:

- **Understanding the importance of data granularity (grain)** before designing analytical views and KPIs.
- **Designing reusable SQL views** instead of writing one-time queries for each business question.
- **Developing business-focused KPIs** that measure revenue, profitability, customer performance, and product returns.
- **Applying window functions** such as `DENSE_RANK()` to support business ranking and performance analysis.
- **Building accurate aggregated metrics**, ensuring calculations such as return rates are performed at the correct level of aggregation.
- **Structuring a reporting layer** that can be directly integrated with business intelligence tools like Power BI.
- **Approaching SQL from a business perspective**, focusing on solving business problems rather than simply retrieving data.

14. Future Improvements

The current project establishes a reusable SQL reporting layer for retail analytics. Future enhancements could include:

- Develop an interactive Power BI dashboard for business users.
- Introduce time-based sales and profitability trend analysis.
- Implement customer segmentation techniques to better understand purchasing behavior.
- Expand the KPI framework by incorporating metrics such as Profit Margin, Average Order Value, and Customer Lifetime Value.

- Integrate additional business data, such as inventory and supplier information, to support more comprehensive retail analytics.

15. Final Architecture

Introduction

The project follows a layered analytical architecture that transforms raw transactional retail data into reusable SQL reporting views. These views serve as the foundation for business analysis and future Power BI dashboards, enabling efficient and scalable business intelligence reporting.

Customers

Orders

Products

Returns

Categories

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SQL Reporting Layer

(vw_product_performance)

(vw_category_performance)

(vw_customer_performance)

(vw_return_details)

|



Business KPIs

Revenue

Profit

Return Rate

Rankings

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Power BI Dashboard

The layered architecture separates raw transactional data from analytical reporting, making the solution reusable, scalable, and easier to maintain. By organizing business logic into dedicated SQL views, the reporting layer provides a reliable foundation for dashboard development, business analysis, and future analytical enhancements.